Explainable Machine-Learning Pipeline for Post-Cholecystectomy Syndrome Risk Prediction: Expanded Manuscript Experiment Report

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Abstract

This expanded report summarizes the current manuscript-oriented experimental package for predicting post-cholecystectomy syndrome (PCS) using structured clinical data. In addition to the original cross-validated machine-learning evaluation and SHAP interpretation, the report now includes conventional clinical statistics, univariate and multivariable logistic regression, risk stratification, bootstrap internal validation, feature-domain ablation, clinical threshold decision tables, calibration comparison, SHAP stability, sensitivity analyses, subgroup performance, and exploratory PCS onset analysis.

1 Background

Post-cholecystectomy syndrome (PCS) may occur after cholecystectomy and can manifest as abdominal pain, dyspepsia, constipation, or other gastrointestinal symptoms. This demo presents a concise workflow for PCS risk prediction, model evaluation, explainability, and exploratory phenotype analysis.

2 Data and Study Design

The dataset included 311 patients. The binary outcome was PCS occurrence, with 70 PCS-positive cases and 241 PCS-negative cases. Candidate predictors included demographics, lifestyle factors, preoperative symptoms, medical history, imaging findings, and laboratory markers. Identifier variables and post-outcome fields were excluded from the binary PCS prediction model.

3 Analytical Pipeline

Numeric variables were imputed using the median and standardized. Categorical variables were imputed using the most frequent category and one-hot encoded. Models were evaluated using stratified five-fold cross-validation. Out-of-fold predictions were used for receiver operating characteristic (ROC) curves, calibration curves, decision curve analysis, and summary metrics.

4 Clinical Baseline and Conventional Statistics

Table 1 provides a condensed version of the formal baseline comparison. The complete table was exported to `output/manuscript_experiments/tables/baseline_table.csv` and `manuscript_experiment_results`

Continuous variables are summarized as median (interquartile range), and categorical variables are summarized as counts and percentages. The strongest group differences were concentrated in hepatobiliary laboratory markers and symptom-related variables.

Table 1: Condensed baseline comparison between PCS and non-PCS groups.

Characteristic	PCS	No PCS	P	FDR
Age	64.00 (48.25-72.00)	59.00 (49.00-67.00)	0.056	0.167
Sex: female	32 (45.7%)	133 (55.2%)	0.175	0.312
Sex: male	38 (54.3%)	108 (44.8%)	0.175	0.312
BMI	22.88 (20.80-24.91)	24.03 (21.72-26.12)	0.037	0.116
Symptom duration: long	1 (1.4%)	27 (11.2%)	¡0.001	¡0.001
Symptom duration: medium	7 (10.0%)	32 (13.3%)	¡0.001	¡0.001
Symptom duration: none	6 (8.6%)	48 (19.9%)	¡0.001	¡0.001
Symptom duration: short	9 (12.9%)	54 (22.4%)	¡0.001	¡0.001
Symptom duration: very_short	47 (67.1%)	80 (33.2%)	¡0.001	¡0.001
Abdominal pain: 0	6 (8.6%)	48 (19.9%)	0.031	0.108
Abdominal pain: 1	64 (91.4%)	193 (80.1%)	0.031	0.108
Pain frequency: daily	48 (68.6%)	92 (38.2%)	¡0.001	0.001
Pain frequency: monthly	7 (10.0%)	55 (22.8%)	¡0.001	0.001
Pain frequency: none	6 (8.6%)	48 (19.9%)	¡0.001	0.001
Pain frequency: occasional	1 (1.4%)	10 (4.1%)	¡0.001	0.001
Pain frequency: weekly	8 (11.4%)	36 (14.9%)	¡0.001	0.001
Stone location: cystic_duct	1 (1.4%)	2 (0.8%)	0.110	0.241
Stone location: distal_common_bile_duct	6 (8.6%)	5 (2.1%)	0.110	0.241
Stone location: gallbladder_body	47 (67.1%)	191 (79.3%)	0.110	0.241
Stone location: gallbladder_fundus	8 (11.4%)	16 (6.6%)	0.110	0.241
Stone location: gallbladder_neck	1 (1.4%)	7 (2.9%)	0.110	0.241
Stone location: left_intrahepatic_bile_duct	0 (0.0%)	1 (0.4%)	0.110	0.241
Stone location: none	7 (10.0%)	19 (7.9%)	0.110	0.241
CBD diameter	6.00 (6.00-7.00)	6.00 (5.00-6.00)	0.035	0.116

5 Logistic Regression and Clinical Statistical Backbone

To complement the machine-learning models with conventional clinical inference, univariate and multivariable logistic regression analyses were performed. The univariate analysis screened candidate associations, while the multivariable model used a reduced clinically interpretable predictor set with variance inflation factor checks to limit collinearity. Results support the same broad signal seen in SHAP: hepatobiliary laboratory abnormalities, especially GGT and CA19-9, were strongly associated with PCS occurrence.

Table 2: Top univariate logistic regression results.

Variable	Term	OR (95% CI)	P	FDR
GGT	ggt	12.90 (6.88-24.20)	¡0.001	¡0.001
ALP	alp	6.28 (3.82-10.34)	¡0.001	¡0.001
ALT	alt	20.30 (8.24-50.01)	¡0.001	¡0.001
Total bilirubin	total_bilirubin	5.52 (3.12-9.75)	¡0.001	¡0.001
AST	ast	45.45 (12.29-168.12)	¡0.001	¡0.001
Total bile acid	total_bile_acid	54.17 (7.66-383.14)	¡0.001	¡0.001
CA19-9	ca199	427.00 (19.28-9456.73)	¡0.001	¡0.001
Pain frequency	pain_frequency_monthly	0.24 (0.10-0.58)	0.001	0.008
Pain frequency	pain_frequency_none	0.24 (0.10-0.60)	0.002	0.012
Symptom duration	symptom_duration_very_short	15.86 (2.09-120.56)	0.008	0.036

Table 3: Multivariable logistic regression after collinearity-aware predictor reduction.

Predictor	Adjusted OR (95% CI)	P
ggt	7.11 (3.31-15.28)	¡0.001
ca199	116.72 (3.75-3630.61)	0.007
total_bilirubin	1.81 (0.98-3.34)	0.056
total_bile_acid	9.56 (0.34-271.84)	0.186
symptom_duration_medium	6.07 (0.41-88.85)	0.188
pain_frequency_weekly	4.31 (0.31-60.73)	0.279
symptom_duration_very_short	6.04 (0.15-240.34)	0.338
pain_frequency_occasional	4.31 (0.16-117.00)	0.385
pain_frequency_none	3.15 (0.07-151.09)	0.561
pain_frequency_monthly	1.17 (0.08-17.00)	0.908
age	0.98 (0.64-1.51)	0.928
symptom_duration_short	1.09 (0.05-23.89)	0.956

6 Model Development and Evaluation

We compared logistic regression, random forest, gradient boosting, support vector machine, Gaussian naive Bayes, XGBoost, and three ensemble architectures. Table 4 summarizes the current cross-validated performance. Random forest achieved the highest out-of-fold area under the ROC curve (AUC).

Table 4: Cross-validated performance for PCS occurrence prediction.

Model	OOF AUC	OOF AP	Brier	Sensitivity	Specificity	F1
Base models						
Random Forest	0.917	0.842	0.088	0.714	0.946	0.747
Gradient Boosting	0.895	0.827	0.081	0.729	0.950	0.768
XGBoost	0.889	0.799	0.091	0.757	0.930	0.756
SVM	0.856	0.799	0.089	0.657	0.967	0.741
Gaussian NB	0.853	0.716	0.459	0.900	0.398	0.462
Logistic Regression	0.849	0.817	0.110	0.743	0.900	0.713
Ensemble models						
Weighted Soft Voting Ensemble	0.915	0.852	0.083	0.729	0.950	0.763
Stacking Ensemble	0.904	0.840	0.106	0.786	0.913	0.752
Soft Voting Ensemble	0.897	0.842	0.080	0.714	0.950	0.758
Hard Voting Ensemble	0.877	0.750	0.086	0.729	0.954	0.773

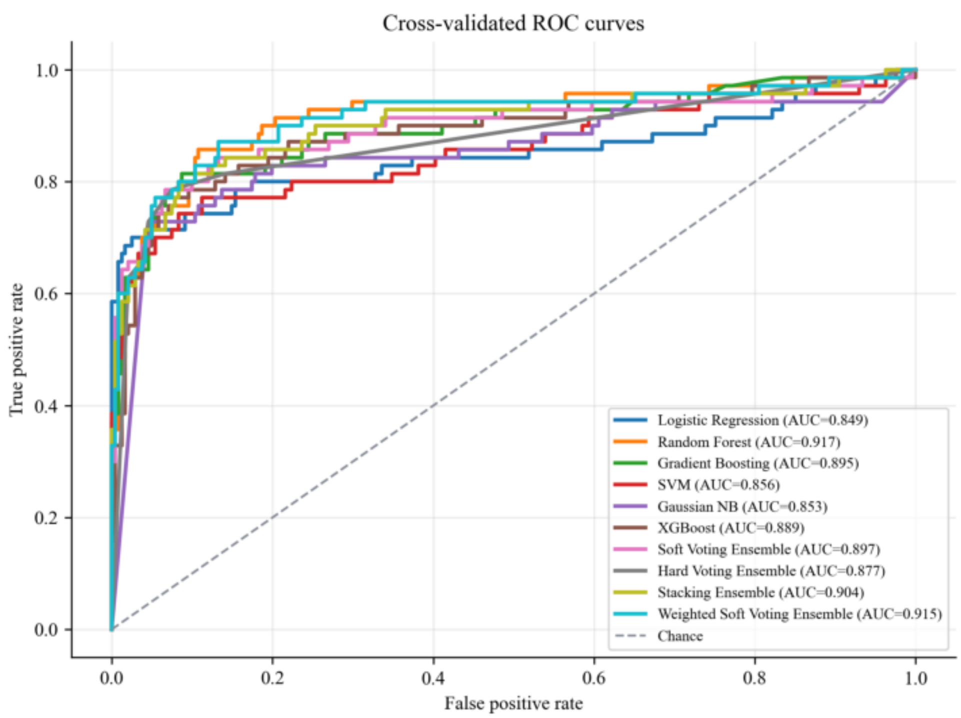
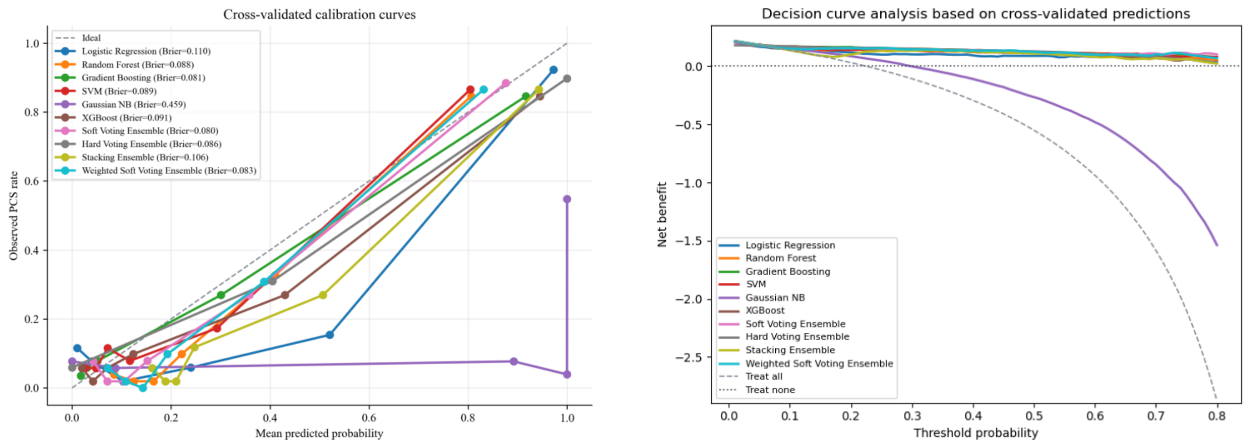


Figure 1: Cross-validated ROC curves based on out-of-fold predictions.



(a) Calibration curves (b) Decision curve analysis

Figure 2: Clinical model assessment beyond discrimination.

7 Ensemble Architecture Validation

To test whether a multi-classifier architecture improved performance, we evaluated soft voting, hard voting, stacking, and an optimized weighted soft-voting ensemble using the same out-of-fold validation protocol. Weighted soft voting selected model weights inside each outer training fold using inner cross-validation. The learned weights consistently favored Random Forest, with smaller contributions from the other base models. The weighted ensemble approached but did not exceed the best single Random Forest model in discrimination (OOF AUC 0.915 vs. 0.917), while improving average precision and Brier score. Thus, ensemble learning did not replace the best single model in this dataset, but weighted soft voting provided a useful architecture check when calibrated risk probabilities are prioritized.

8 Risk Stratification and Clinical Thresholds

Random Forest out-of-fold probabilities were converted into clinically interpretable low-, intermediate-, and high-risk groups. The high-risk group captured most PCS-positive patients and had a markedly higher observed PCS rate than the low- and intermediate-risk groups. This analysis translates the model from discrimination metrics into a candidate follow-up allocation tool.

Table 5: Observed PCS rate by model-defined risk strata.

Risk group	N	PCS cases	Observed PCS rate	Mean predicted risk
Low (¡10%)	42	2	4.8%	8.1%
Intermediate (10-30%)	175	8	4.6%	17.3%
High (¿=30%)	94	60	63.8%	64.3%

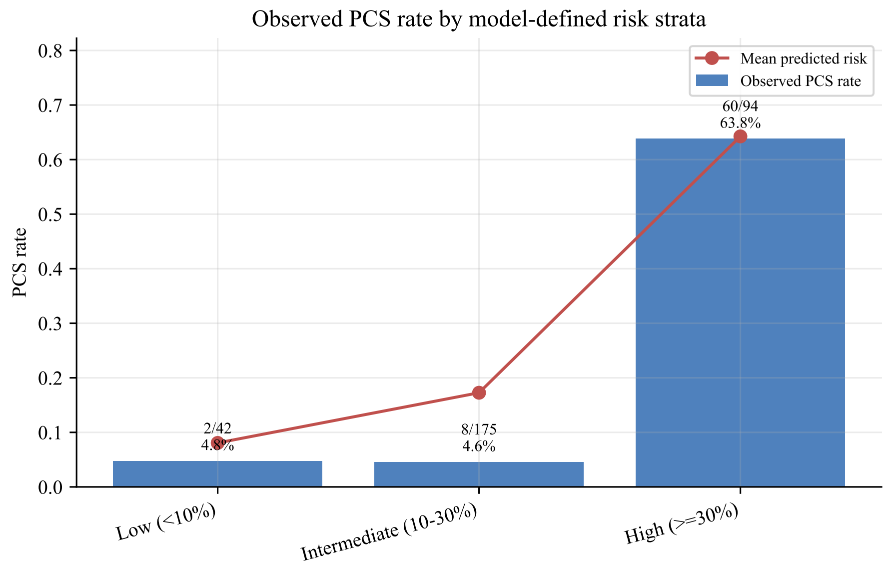


Figure 3: Observed PCS rate and mean predicted risk across low-, intermediate-, and high-risk strata.

Clinical threshold analysis was performed at 10%, 20%, and 30% predicted PCS risk. At the 30% threshold, the model identified 60 of 70 PCS cases while reducing false positives compared with lower thresholds.

Table 6: Clinical threshold decision table based on out-of-fold predicted probabilities.

Threshold	Sensitivity	Specificity	PPV	TP/100	FP/100	FN/100
10%	0.971	0.166	0.253	21.9	64.6	0.6
20%	0.943	0.660	0.446	21.2	26.4	1.3
30%	0.857	0.859	0.638	19.3	10.9	3.2

9 Bootstrap Internal Validation

To address overfitting concerns in a dataset with 70 PCS-positive cases, bootstrap internal validation was performed with 500 resamples. The optimism-corrected AUC remained high, supporting the stability of the discrimination signal, while calibration slope results indicate that probability calibration still requires caution and external validation.

Table 7: Bootstrap internal validation of the Random Forest model.

Metric	OOF apparent	Bootstrap mean	CI low	CI high	Optimism-corrected
auc	0.917	0.969	0.951	0.985	0.889
brier	0.088	0.059	0.055	0.064	0.111
calibration_intercept	-0.324	-0.101	-0.547	0.449	-0.288
calibration_slope	1.610	2.163	1.610	2.855	0.253
prediction_instability_sd	NA	0.057	0.023	0.173	0.044

10 Feature-Domain Ablation

Feature-domain ablation was used to test whether multi-source clinical data improved predictive performance beyond any single feature domain. Laboratory-only and symptoms-plus-laboratory models performed strongly, indicating that hepatobiliary laboratory markers carried the dominant signal in this cohort. The all-variable model remained clinically attractive because it preserves broader context and interpretability, but these results make clear that laboratory features are central to PCS prediction.

Table 8: Feature-domain ablation experiment.

Feature set	AUC	AP	Brier	Sensitivity	Specificity
Symptoms + Laboratory	0.925	0.846	0.084	0.714	0.950
Laboratory	0.923	0.835	0.082	0.743	0.938
All variables	0.913	0.842	0.089	0.714	0.946
Symptoms	0.638	0.287	0.224	0.657	0.660
Demographics	0.566	0.307	0.211	0.300	0.797
Imaging	0.563	0.308	0.228	0.386	0.726

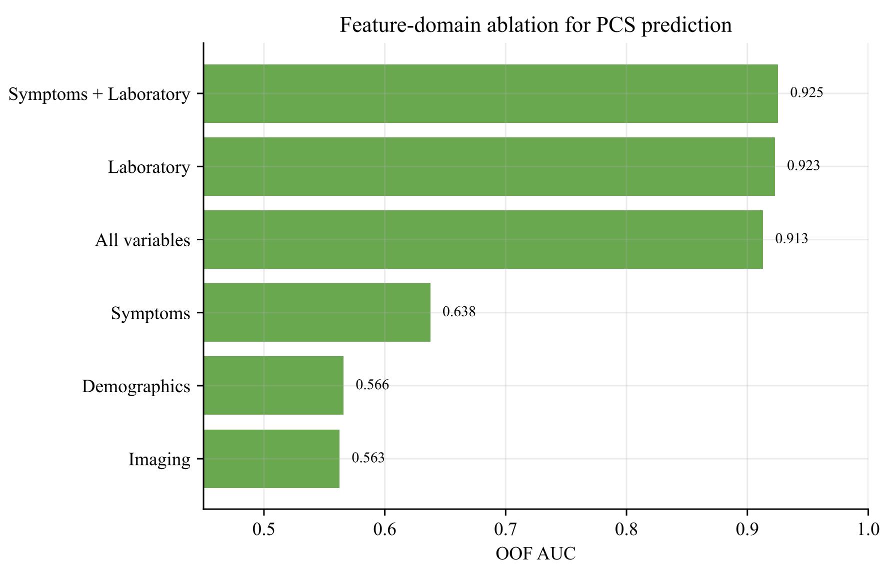


Figure 4: AUC comparison across feature-domain ablation models.

11 Calibration Enhancement

Because clinical prediction models require usable probabilities rather than discrimination alone, raw Random Forest probabilities were compared with Platt scaling and isotonic calibration. Platt scaling achieved the lowest Brier score and a calibration slope closer to 1 in the current internal validation.

Table 9: Calibration method comparison.

Calibration method	AUC	Brier	Intercept	Slope
Platt scaling	0.911	0.081	-0.016	1.077
Isotonic calibration	0.908	0.084	-0.516	0.630
Raw RF	0.913	0.089	-0.339	1.585

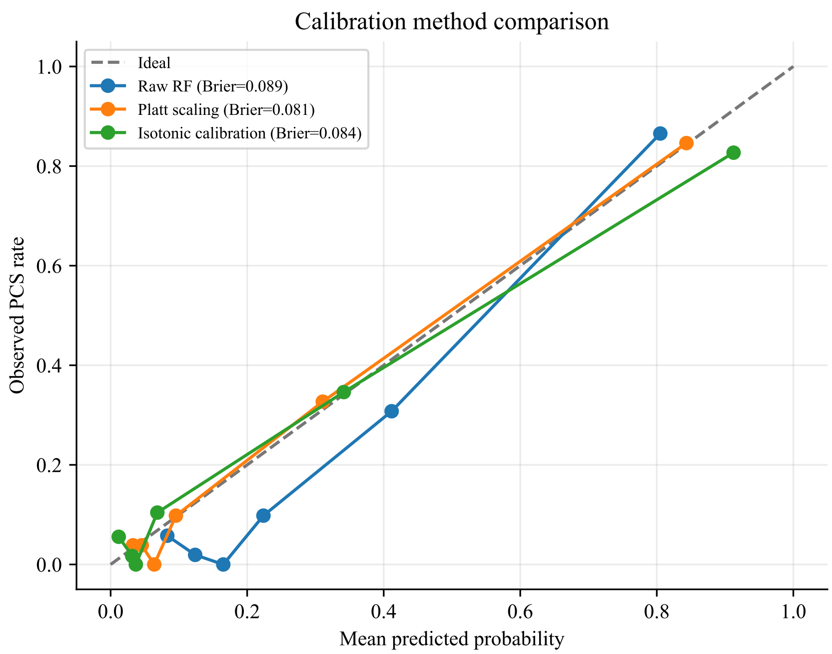


Figure 5: Calibration curves for raw Random Forest, Platt scaling, and isotonic calibration.

12 SHAP-Based Interpretation

Random forest was selected for SHAP analysis because it achieved the best out-of-fold AUC. The fitted model mainly relied on hepatobiliary laboratory markers and symptom-related variables. The leading variables by aggregated mean absolute SHAP value were GGT, ALT, AST, ALP, CA19-9, total bilirubin, symptom duration, total bile acid, pain frequency, and age.

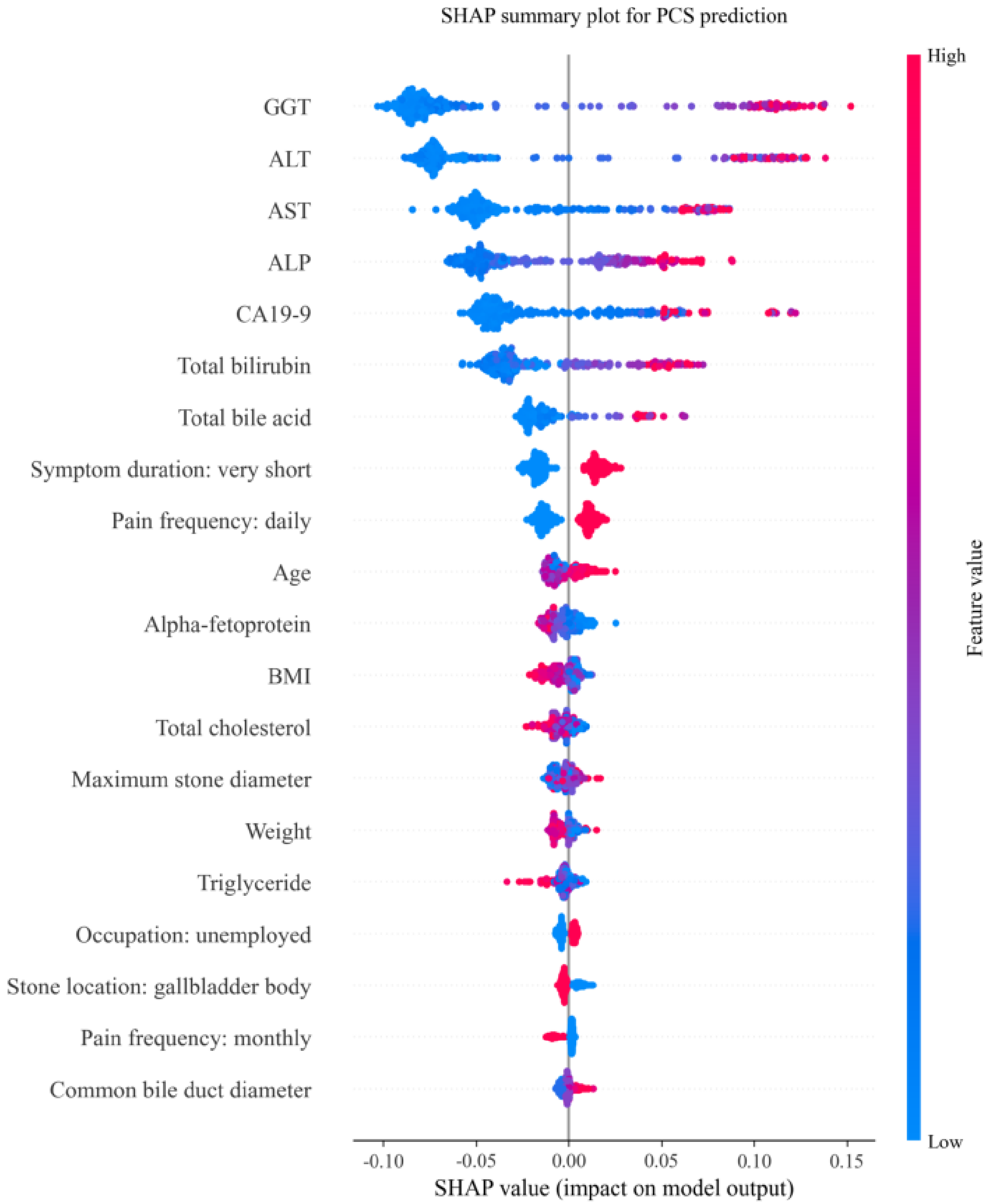


Figure 6: SHAP summary plot for the final random forest model.

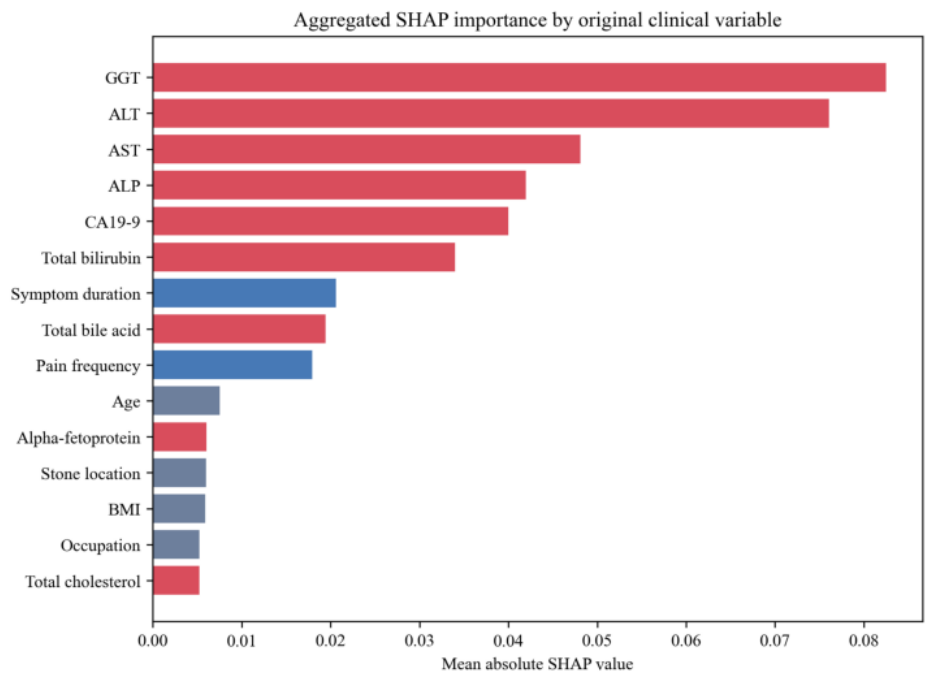


Figure 7: Aggregated SHAP importance by original clinical variable.

13 SHAP Stability

To test whether the explanatory pattern was fold-specific, SHAP rankings were recalculated across cross-validation folds. GGT, ALT, ALP, AST, CA19-9, total bilirubin, symptom duration, and total bile acid appeared in the top 10 features in all folds, supporting the robustness of the model explanation.

Table 10: Cross-fold SHAP top-10 stability.

Feature	Top-10 frequency	Folds	Mean rank
GGT	100.0%	5	1.4
ALT	100.0%	5	1.6
ALP	100.0%	5	3.4
AST	100.0%	5	4.2
CA19-9	100.0%	5	5.0
Total bilirubin	100.0%	5	5.4
Symptom duration	100.0%	5	7.6
Total bile acid	100.0%	5	7.6
Pain frequency	80.0%	4	8.8
Stone location	40.0%	2	10.0

14 Sensitivity and Subgroup Analyses

Sensitivity analyses assessed whether the main model signal persisted under altered preprocessing and class-weight assumptions. The model remained stable after numeric winsorization and mean imputation; removing class weighting increased specificity but reduced sensitivity. Subgroup analyses showed generally consistent discrimination across sex, age, and abdominal-pain strata, although small subgroups require cautious interpretation.

Table 11: Sensitivity analyses for preprocessing and class weighting.

Analysis	AUC	Brier	Sensitivity	Specificity
No class weight	0.914	0.081	0.657	0.971
Base RF	0.913	0.089	0.714	0.946
Winsorized numeric 1-99%	0.913	0.089	0.714	0.946
Mean numeric imputation	0.911	0.090	0.714	0.946

Table 12: Subgroup performance using Random Forest out-of-fold predictions.

Subgroup	N	PCS	AUC	Brier	Sensitivity	Specificity
No abdominal pain	54	6	0.951	0.059	0.500	1.000
High GGT	158	61	0.920	0.116	0.803	0.866
Age ¿=60	157	40	0.917	0.101	0.700	0.923
Female	165	32	0.917	0.081	0.656	0.970
Male	146	38	0.915	0.097	0.763	0.917
Age ¡60	154	30	0.913	0.075	0.733	0.968
Abdominal pain	257	64	0.911	0.094	0.734	0.933
Low GGT	152	8	0.669	0.060	0.000	1.000

15 Exploratory PCS Onset Analysis

PCS onset time was not used as a predictor. As an exploratory descriptive analysis among PCS-positive patients with available onset records, high-risk patients had a shorter median onset time than lower-risk groups. However, the Spearman correlation between predicted risk and onset days was weak, and this analysis cannot be interpreted as a formal time-to-event model because censoring and follow-up windows are not fully specified.

Table 13: Exploratory PCS onset summary by predicted risk group.

Risk group	N	Median onset/rho	IQR/p	Mean risk
Low (¡10%)	2	91.5	59.5	0.079
Intermediate (10-30%)	8	76.5	65.2	0.218
High (¿=30%)	59	45.0	82.5	0.729
Spearman risk vs onset	69	-0.007	0.956	0.651

16 Exploratory PCS Symptom-Category Task

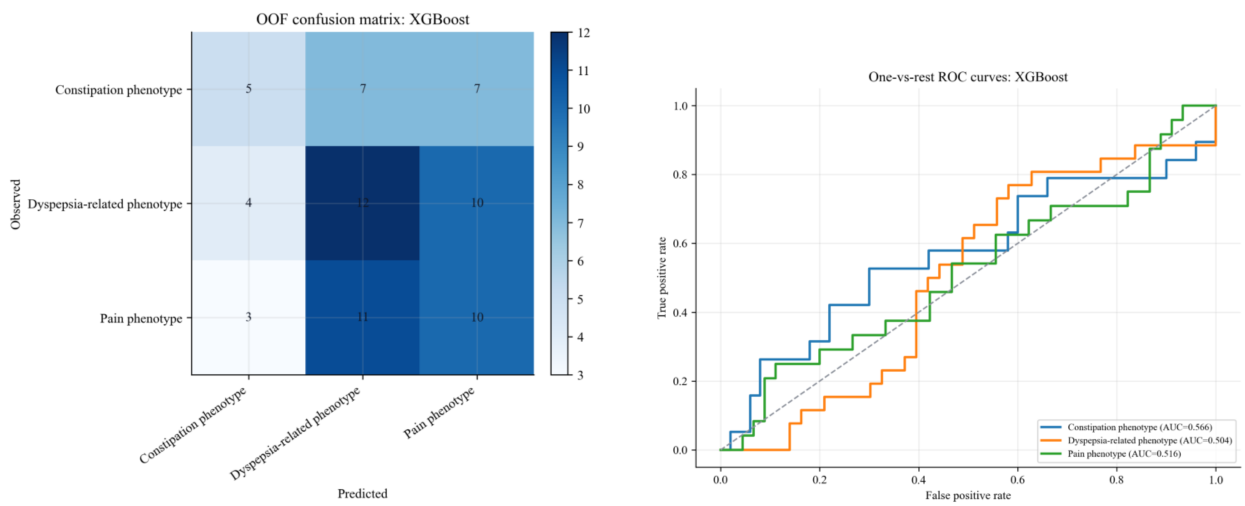
The original free-text field for PCS symptom type was mapped into structured labels. This downstream task was restricted to PCS-positive patients with known symptom text. PCS-negative patients were not included in the symptom-subtype classifier. PCS-positive symptom text was grouped into pain, constipation, and dyspepsia-related phenotypes. Raw entries originally recorded as intestinal dysbiosis or gastrointestinal dysfunction were treated as dyspepsia symptoms for labeling, while the underlying mechanisms can be discussed separately. This task was treated as an exploratory secondary analysis because symptom subtype sample sizes were small.

Table 14: Constructed PCS symptom occurrence categories in the full cohort.

Category	Description	Count
`no_pcs`	No PCS	241
`pain`	Abdominal pain phenotype	24
`constipation`	Constipation phenotype	19
`gi_dysfunction_dyspepsia`	Dyspepsia-related phenotype	26
`pcs_unknown`	PCS-positive with missing symptom text	1

Table 15: Exploratory downstream symptom-subtype prediction performance among PCS-positive patients.

Model	Accuracy	Balanced accuracy	Macro F1	Macro OVR AUC
XGBoost	0.391	0.380	0.381	0.529
Multinomial Logistic	0.391	0.382	0.381	0.565
Random Forest	0.391	0.373	0.365	0.548
Gradient Boosting	0.362	0.351	0.348	0.544
SVM	0.246	0.225	0.197	0.445



(a) Confusion matrix

(b) One-vs-rest ROC

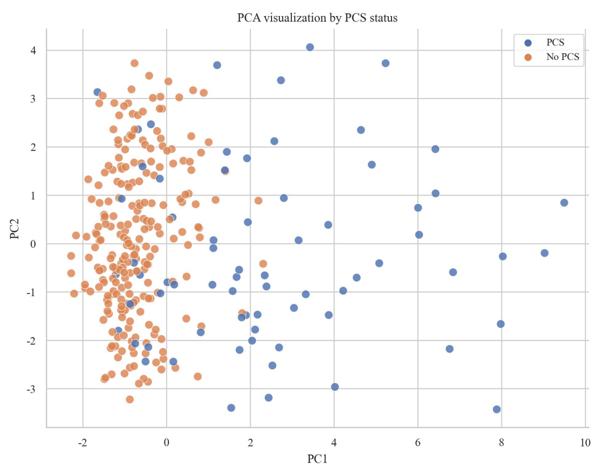
Figure 8: Exploratory downstream PCS symptom-subtype prediction results. Balanced accuracy is calculated only across PCS symptom subtypes and excludes `no_pcs`.

17 Exploratory Latent Phenotype Analysis

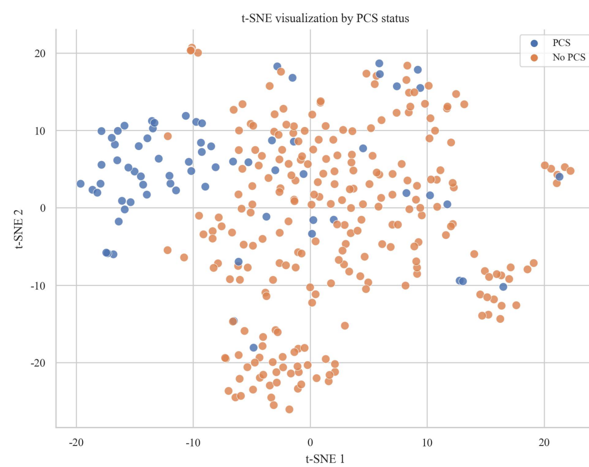
Unsupervised dimensionality reduction and clustering were used to explore whether the preoperative feature space contained latent phenotypes related to PCS occurrence or symptom subtypes. This analysis was exploratory and hypothesis-generating. K-means selected a two-cluster solution based on the highest silhouette score. Cluster 0 showed a markedly higher PCS rate than cluster 1.

Table 16: Latent cluster summary.

Cluster	N	PCS cases	PCS rate	Median PCS onset, days
0	45	43	95.6%	43.0
1	266	27	10.2%	86.0

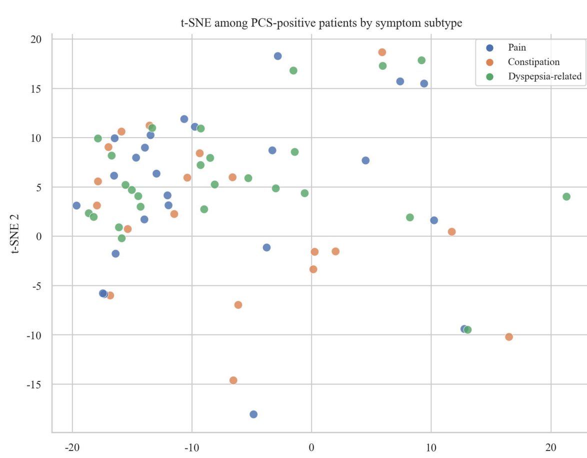


(a) PCA by PCS status

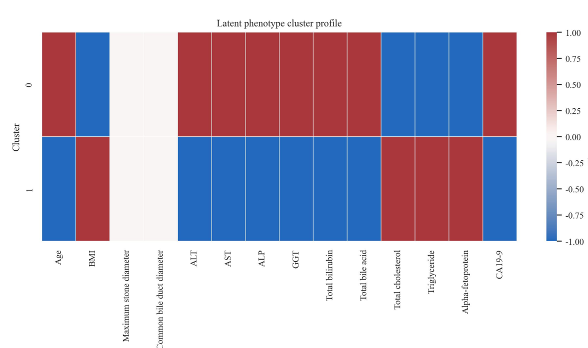


(b) t-SNE by PCS status

Figure 9: Exploratory dimensionality reduction using preoperative variables.



(a) PCS-positive patients by symptom subtype



(b) Cluster feature profile

Figure 10: Exploratory latent phenotype structure.

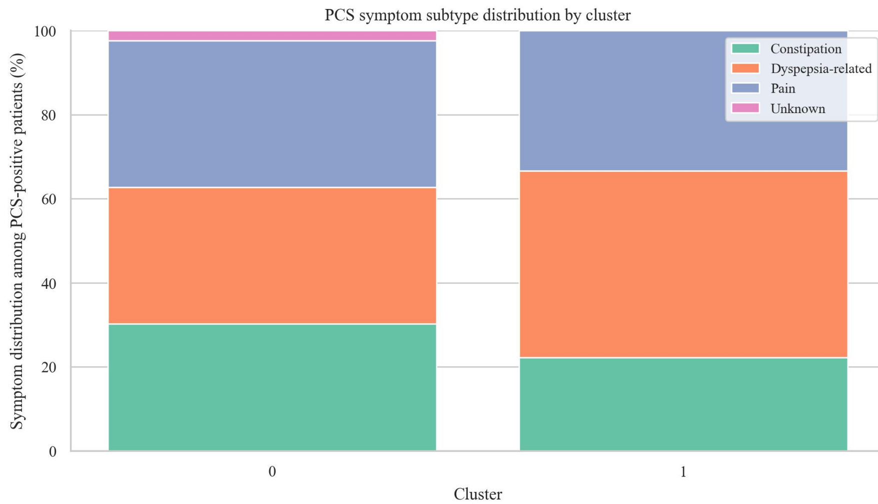


Figure 11: PCS symptom subtype distribution by latent cluster. Cluster-level PCS rates are reported in Table 16.

18 Current Interpretation

The expanded results suggest that machine-learning models can provide strong internal discrimination for PCS occurrence, with random forest achieving the best out-of-fold AUC among the original model set. Conventional logistic regression, feature ablation, and SHAP analyses consistently point to hepatobiliary laboratory markers as the main prediction signal. Risk stratification and threshold tables make the model more clinically interpretable by showing how predicted probabilities could inform follow-up intensity. The symptom-category and latent phenotype tasks provide an initial framework for modeling PCS heterogeneity, but their class sizes and exploratory design require cautious interpretation.

19 Limitations

This report uses internal validation only. Although bootstrap validation, sensitivity analyses, calibration comparison, and feature ablation improve the evidence base, external validation and prospective assessment remain necessary before clinical deployment. The cohort is small, PCS-positive cases are limited, and the study lacks a fully specified censoring framework for formal survival analysis. Symptom subtype prediction and latent phenotype analysis should be treated as exploratory.